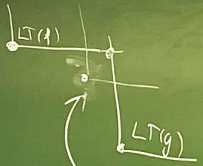
2.5.2. Definition of $S(f,g)$ for $f,g \in k[x_1,\dots,x_n]$

2.5.2'. Remark



Intention: with $f,g \in I$ (some ideal) to obtain $LT(\underbrace{S(f,g)}_{\in I})$ that is not yet "covered"

2.5.5 Theorem (S-pair criterion). For $g_1,\dots,g_s \in k[x_1,\dots,x_n]\setminus\{0\}$, the following are equivalent.

(i) $g_1,\dots,g_s$ is a Gröbner basis of $I := \langle g_1,\dots,g_s\rangle$

(ii) for all $1 \leq i,j \leq s$ the remainder of the division of $S(g_i,g_j)$ by $g_1,\dots,g_s$ is zero.

Proof: see lecture notes.

$\boxed{2.6}$ Buchberger's algorithm.

2.6.1 Theorem. There exists an algorithm that takes polynomials $f_1,\dots,f_s \in k[x_1,\dots,x_n]$ as an input and generates a Gröbner basis of $\langle f_1,\dots,f_s\rangle$.

Proof: Gröbner_basis$(f_1, \dots, f_s)$ ▷ Let's assume $f_i \neq 0$ $(i = 1, \dots, s)$.

$G := (f_1, \dots, f_s)$ ▷ Gröbner basis, under construction

while True:

  $R := ()$ ▷ list of new polynomials, under construction

  for all $p, q \in G$ with $p \neq q$:

    determine the remainder r of the division of $S(p,q)$ by G

    if $r \neq 0$, add r to R.

  end

  if R is an empty list, return G ▷ termination.

  expand G by adding elements of R to it ▷ the case R is not empty

end

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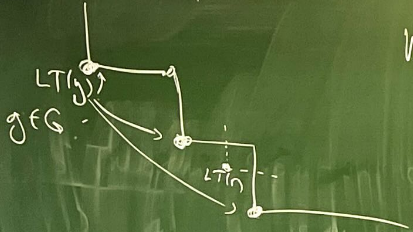
If this algorithm terminates, it gives a Gröbner basis of $\langle f_1, \ldots, f_s \rangle$, by Theorem 2.5.5, because it only adds polynomials from $I = \langle f_1, \ldots, f_s \rangle$ to G and when terminates, cond. (ii) in Th. 2.5.5 is satisfied. Let's discuss why it terminates.



When r is added to R, then $LT(r) \notin \langle LT(g) : g \in G \rangle$ so that after adding the polynomials $r \in R$ to G, the monomial ideal $\langle LT(g) : g \in G \rangle$ grows strictly.

But the ascending chain condition, an ascending chain of ideals stabilizes (stop growing at some point). So the growth will not go on forever. $\square$

2.6.2. Implementation in Sage Math (see lecture notes).

<u>2.6.3.</u> Example $I = \langle \underbrace{x^2 - 2xy}_{g_1}, \underbrace{x^2y - 2y^2 + x}_{g_2} \rangle$

Let's calculate a Gröbner basis w.r.t. the <u>lex</u> order.

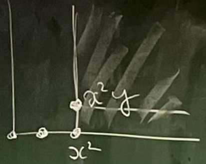
$$S(g_1, g_2) = y g_1 - g_2 = (x^2y - 2xy^2) - (x^2y - 2y^2 + x)$$
$$= -2xy^2 + 2y^2 - x$$

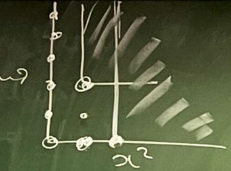
We divide $S(g_1, g_2)$ by g_1, g_2 and, if the result is non-zero, this is our g_3.

to be divided	remainder under constr	g_1	g_2
$-2xy^2 + 2y^2 - x$	0	0	0
$2y^2 - x$	$-2xy^2$	0	0
$2y^2$	$-2xy^2 - x$	0	0
			0
	$-2xy^2 - x + 2y^2$	0	0

$\parallel$

g_3 (or we can rescale it)





$$g_1 = x^2 - 2xy$$
$$g_2 = x^2y - 2y^2 + x$$
$$g_3 = xy^2 + \tfrac{1}{2}x - y^2$$

$S(g_1,g_2) \rightsquigarrow S(g_2,g_3)$, and $S(g_1,g_3)$ to be divided by (g_1,g_2,g_3)

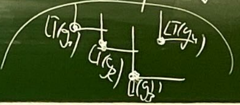
......

And, if our order is different, the Gröbner basis will be different.

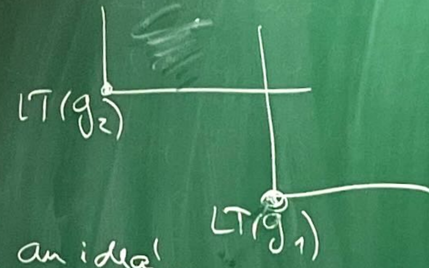
SageMath can do it for you (also other computer algebra systems).

Actually, for every given monomial order, one introduce *the* Gröbner basis of an ideal $I \subseteq k[x_1,\dots,x_n]$. What's the way to do this?

- One can rescale the polynomials to have the leading coefficient equal 1.
- We can remove the polynomials whose leading term doesn't contribute to the initial ideal:
  if $g_1,\dots,g_s$ is a Gröbner basis of I and $LT(g_i) \in \langle LT(g_1),\dots,LT(g_{i-1}), LT(g_{i+1}),\dots,LT(g_s)\rangle$, then $g_1,\dots,g_{i-1},g_{i+1},\dots,g_s$ is still a Gröbner basis of I.



- You can reduce your polynomials so that no term of g_i is divisible by the $LT(g_j)$ for any other $j \neq i$.



If you have all those three properties, your Gröbner basis is unique as a set.
This is called the reduced Gröbner basis of an ideal!

Remark/Example related to linear algebra. Let's calculate the reduced Gröbner basis from polynomial of degree one (polynomials that define a linear system of equations). Let's fix the monomial order lex.

$$\begin{cases} \underline{x_1} + x_2 + x_3 + x_4 - 1 = 0 \\ \underline{x_1} - x_2 + 2x_3 - x_4 - 2 = 0 \end{cases}$$

The Gröbner basis algorithm that calculates the reduced Gröbner will calculate the reduced echelon form of the system.

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$$\begin{cases} \underline{\underline{x_1}} + x_2 + x_3 + x_4 - 1 = 0 \\ -2\underline{\underline{x_2}} + x_3 - 2x_4 - 1 = 0 \end{cases}$$

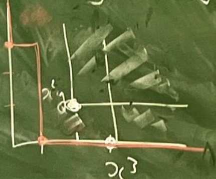
$$\begin{cases} \underline{\underline{x_1}} + x_2 + x_3 + x_4 - 1 = 0 \\ \underline{\underline{x_2}} - \frac{1}{2}x_3 + x_4 + \frac{1}{2} = 0 \end{cases}$$

these two polynomials are a Gröbner basis of $\langle x_1 + x_2 + x_3 + x_4 - 1,\ x_1 - x_2 + 2x_3 - x_4 - 2 \rangle = I$

$$\begin{cases} \underline{\underline{x_1}} \quad + 1\frac{1}{2}x_3 \quad - 1\frac{1}{2} = 0 \\ \underline{\underline{x_2}} - \frac{1}{2}x_3 + x_4 + \frac{1}{2} = 0 \end{cases}$$

these two polynomials form <u>the</u> reduced Gröbner basis of I.

<u>Example</u> $\langle x^3 - 2xy,\ x^2y - 2y^2 + x \rangle$ lex-order.



The reduced Gröbner basis

$\underline{x} - 2y^2$

$\underline{y^3}$

deglex-order

$\underline{\underline{x^2}}$

$\underline{\underline{xy}}$

$\underline{y^2} - \frac{1}{2}x$